

Apply to be a Science Corps Fellow - Entry #3329

Name

FATIMA ANWAR

Email

fatma.anwar74@gmail.com

Gender

Female

Residence: City (State), Country

Dallas Texas USA

Nationality

Pakistani

When did you or do you expect to complete your PhD?

University of North Texas

Please tell us briefly about your university education (institution, city, country, year)

****Ph.D. in Analytical Chemistry | University of North Texas | Dec 2023****

My Ph.D. research focused on **Surface Engineering of Materials for Beyond Microelectronics**, exploring advanced materials for renewable energy, optoelectronics, and environmental applications. I specialized in ****Thin Film Technology****, ****Surface Science****, ****Semiconductors****, ****Electrocatalysis****, and ****Optical Devices****, developing materials with enhanced conductivity, catalytic activity, and light-matter interactions. My work involved techniques like CVD and ALD, aiming to overcome limitations in traditional materials.

I also gained expertise in teaching, mentoring, guiding students in chemistry courses and laboratory research. My research resulted in peer-reviewed publications and conference presentations, contributing to the scientific community.

Moving forward, I aim to leverage my skills in research and education to inspire future innovators and address global challenges through STEM advancement.

Why would you like to be a Science Corps Fellow?

Aligning with Science Corps' Mission:

Building Global STEM Capacity Through Passion and Purpose

Science Corps' mission to build global STEM capacity aligns with my professional values and personal passion. As an international student from Pakistan, I've experienced the transformative power of quality education. My journey

from a developing nation to earning a PhD from a world-class institution has inspired me to give back, particularly to regions lacking access to quality STEM education. Pakistan, like many developing countries, faces challenges such as outdated curricula, limited resources, and socioeconomic barriers that discourage young people from pursuing STEM careers.

Globally, fewer than 30% of researchers are women, with even fewer in developing countries. Initiatives like Science Corps are vital in bridging this gap by fostering a global community of innovators. My journey exposed me to cutting-edge research and collaborative environments, deepening my understanding of analytical chemistry and its role in solving global challenges.

I am committed to addressing disparities in STEM education by leveraging my knowledge and experience. By mentoring, sharing relatable success stories, and promoting innovative teaching methods, I aim to inspire young people to pursue STEM fields and contribute to sustainable development. Science Corps' mission resonates with my goal of creating equitable opportunities and empowering the next generation of scientists and engineers. Together, we can inspire young minds to tackle global challenges and drive meaningful change.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

Teaching and Mentoring Experience: A Foundation for Science Corps' Mission

Over the course of my academic career, I have accumulated extensive teaching and mentoring experience, which I believe positions me well to contribute to the mission of Science Corps. During my PhD at the University of North Texas (UNT), I served as a Graduate Assistant for five years, teaching **General Chemistry** and **Organic Chemistry** courses and labs to undergraduate students. This role allowed me to develop and refine my teaching skills, adapt to diverse learning styles, and foster an engaging and inclusive classroom environment.

Prior to my PhD, I taught **Physical Chemistry** and **Inorganic Chemistry** to undergraduate students in Pakistan, where I gained valuable experience in addressing the unique challenges faced by students in resource-constrained settings. Additionally, I have taught high school chemistry, further broadening my understanding of how to make complex scientific concepts accessible and relatable to younger learners.

My commitment to innovative teaching was recognized through a **Chemistry Education Research Scholarship**, awarded for my efforts to modify coursework for undergraduate students during the COVID-19 pandemic. This experience honed my ability to adapt teaching methods to unprecedented challenges, ensuring students continued to receive quality education despite disruptions.

These diverse teaching and mentoring experiences have equipped me with the skills to inspire and support students from varied backgrounds. I am confident that my ability to connect with learners, adapt to different educational contexts, and promote inclusive learning environments will enable me to effectively contribute to Science Corps' mission of building global STEM capacity. By leveraging my expertise, I aim to empower the next generation of scientists and engineers, particularly in underserved regions, and drive meaningful change through education.

Where do you imagine yourself professionally in the future? Check all that apply.

Doing science in industry

Administrative role in industry or academia

In science outreach

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.****Vision for the Future: Bridging Gaps and Inspiring the Next Generation****

In the near future, I aim to expand my expertise by gaining more exposure to the industrial applications of my academic knowledge in chemistry. I want to bridge the gap between theoretical understanding and real-world problem-solving, particularly in fields like materials science, renewable energy, and environmental sustainability. While pursuing this goal, I am committed to giving back to the global community by sharing my knowledge and experiences with the younger generation. Through platforms like Science Corps, I hope to demonstrate the practical applications of STEM and inspire young minds to pursue careers in science and engineering.

In the long term, my vision is to establish scientific communities in less privileged regions, such as Pakistan, where access to quality education and research opportunities is limited. I aspire to serve as a role model, showing young people the wonders of science and its potential to transform lives. To achieve this, I plan to establish small-scale research and development (R&D) facilities that empower local talent to innovate and contribute to fast-growing scientific fields. These facilities will not only create opportunities for local communities but also foster a culture of curiosity, collaboration, and problem-solving.

Additionally, I aim to bridge the gap between the developed and underdeveloped worlds by serving as a mentor and coach for young scientists. I want to facilitate scientific collaborations, provide guidance, and help aspiring researchers overcome barriers to success. By doing so, I hope to create a more equitable global scientific community where talent from all regions can thrive and contribute to solving the world's most pressing challenges.

Through these efforts, I envision a future where science becomes a powerful tool for empowerment and progress, particularly in underserved regions. My ultimate goal is to inspire the next generation of scientists and engineers, equipping them with the skills, knowledge, and opportunities they need to make a meaningful impact on the world.

Please upload your CV (.pdf) [Fatima-Anwar_Resume_2025.pdf](#)**How did you find out about Science Corps?**

Other

[Science Corps](#)